Choose what must be isolated.

Discovery visibility, data delivery, or security?

Different goals need different controls.

DP = DomainParticipant, the DDS entity that owns a discovery scope.

Seeing a topic is not receiving its data

Mechanism	Discovery effect	User-data effect	Main limit
Publisher/Subscriber partitions	Endpoint metadata can remain visible	Only matching audiences exchange data	Not metadata isolation
DP partitions	Unmatched scopes do not exchange endpoint metadata	No matching across those scopes	Applies to every endpoint on that DP
C2 directory, incoming peers accepted	Supplies discovery contacts	Topic matching and routes still apply	Contact list, not an exclusive allowlist
DDS Security	Authentication/protection per policy	Authorized reads/writes and configured protection	Identity, keys, and permission lifecycle

Takeaway: discovery control, data-audience control, and authorization are separate.

Application policy can drive DDS controls

Policy choice	Who decides?	Who applies it?
Which topic routes exist or forward	Application / operator	Router and endpoint lifecycle
Which peers to contact	C2 / application	DDS discovery APIs: `add_peer`, `remove_peer`
Which participant scopes match	Application / XML / admin	DDS participant matching
Which endpoints exchange data	Application / XML / admin	DDS Publisher/Subscriber matching
Which samples pass a filter	Application / operator	DDS reader filter or application forwarding
Which network paths are reachable	Network controller	Transport/firewall rules
Which identities may read/write	Security authority	DDS security plugins

> **Takeaway:** an API called by an application can still enforce a DDS-level rule.

One WAN participant or two?

Required outcome	Arrangement	Trade-off
Global health/navigation; team-only payloads	One WAN DP; endpoint partitions	Simple lifecycle, broader metadata visibility
Global services; team-only endpoint metadata	Global WAN DP + team WAN DP	Separate discovery scopes, extra DP overhead
One restricted discovery audience for all traffic	One DP with that membership scope	Every endpoint shares that discovery boundary

Counts exclude each platform's LAN participant.

> **Takeaway:** DP partitioning does not inherently require another DP.
> Two DPs help when global and team-only discovery must coexist.

A common `SHARED` participant partition makes platforms eligible to discover one another.

Other controls solve different problems

Control	Benefit	Cost or limitation
Separate domains / tags	Separate DDS matching scopes	Concurrent scopes need participants and deliberate bridges
Content / destination filters	Select samples within a topic	Do not hide peer or endpoint metadata
Paused route	Stops that path's forwarding	Endpoints may still be discoverable
On-demand endpoints	Remove dormant route metadata	Startup/type/match delay on activation
Network segmentation	Blocks unwanted network paths	Must account for discovery, payload, and repair traffic

Takeaway: fewer routes or packets do not automatically mean fewer discovered peers.

Squad takeaway and evidence

- Keep one WAN DP when global visibility and team-only data meet the requirement.
- Use C2 to manage contacts and team policy, not as an implied security boundary.
- Add a second scope only when team metadata must be hidden from global peers.
- Measure bandwidth and convergence; do not assume an extra DP is cheaper.

Tested: isolated C2 add-once bootstrap and documented endpoint team isolation.
Not yet tested: teammate repair and a loaded Routing Service peer-management processor.

Full comparison and evidence

In 7.7.0, `remove_peer` is documented for SPDP only; peer removal is not an authorization design.